

WASP-39b

R-1755

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_130517 · created 2026-08-02T20:19:15+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456429.706 +/- 0.053 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.16 +/- 0.051
Transit depth (Rp/Rs)^2	2.56 +/- 1.62 [%]
Orbital Inclination (inc)	89.67 +/- 1.91
Airmass coefficient 1 (a1)	0.844 +/- 0.019
Airmass coefficient 2 (a2)	0.125 +/- 0.028
Scatter in the residuals of the lightcurve fit is	2.22 %
Best Comparison Star	#2 - [314, 43]
Optimal Aperture	4.18
Optimal Annulus	9.73
Transit Duration (day)	0.037 +/- 0.038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.16 ± 0.051 vs 0.1457 (deviation 0.28σ)
Duration fit vs published	0.037 d vs 0.12 d (deviation 2.18σ)
Mid-time O–C	-113.3 min
Depth SNR	3.1
Residual scatter	2.22 %

Light curve

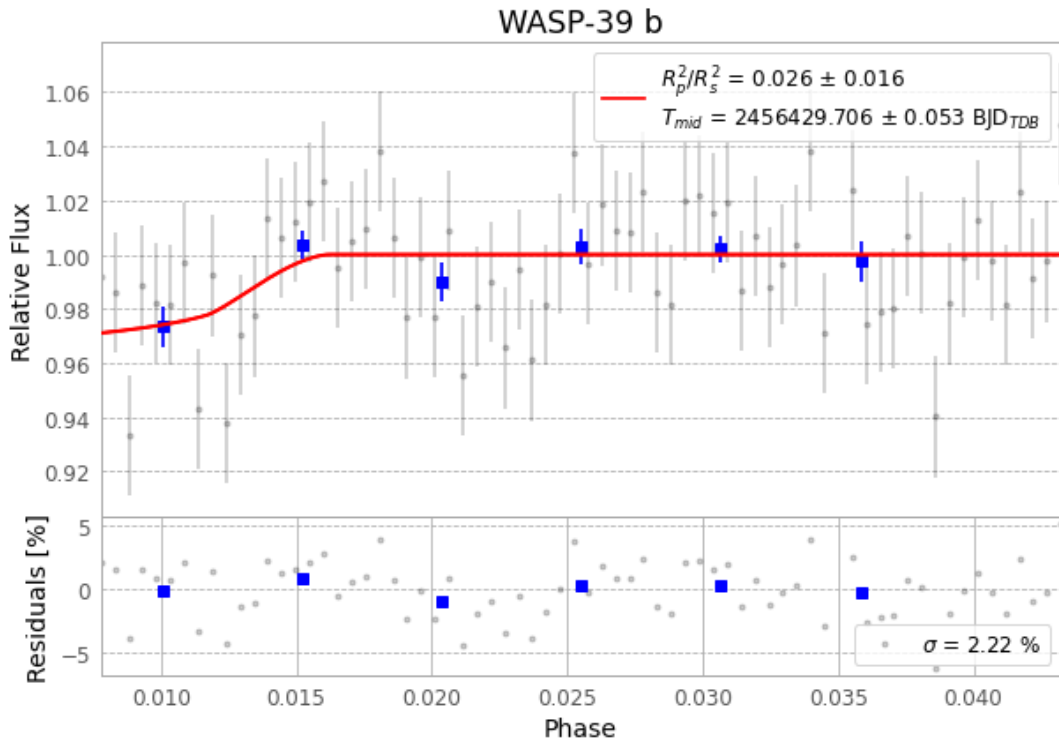


Figure 1 — FinalLightCurve_WASP-39 b_2013-05-16.png (SHA-256 cfae7a03785cde18...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [226, 288], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3b50e7c440e1746e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

71 FITS frames (47 MB), WASP-39130517053012.FITS → WASP-39130517090012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-130517.log (a084d78ec42e...), FinalLightCurve_WASP-39 b_2013-05-16.png (cfae7a03785c...), FinalLightCurve_WASP-39 b_2013-05-16.csv (d1cfa49230ee...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)